Right Answer, Wrong Reason: Hallucinations in AI Explanations
Project Proposal for NLP Course, Winter 2026/27

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Abstract

Large Language Models can often produce correct answers while giving explanations that are partially unsupported, misleading, or factually wrong. This project investigates that phenomenon in chart-based question answering, where both the final answer and the reasoning trace should be grounded in visual evidence. We will compare selected models on chart images and evaluate how often they produce a correct answer with an incorrect or unsupported explanation. Our study combines manual annotation on a small, high-quality sample with larger-scale automatic evaluation, and it examines the influence of prompting strategies on the performance.

1 Introduction

Trustworthiness of LLMs (Large Language Models) outputs is heavily undermined by hallucinations, which are considered one of the most substantial challenges in AI development. Latest models frequently include reasoning techniques that simulate human thinking. There is evidence that these mechanisms reduce the amount of hallucinations in answers. However, recent research suggests that explanations generated by LLMs can contain false data, faulty logic or flawed reasoning steps even if the answer itself is correct. This is a major issue in scientific applications, where the explanation of an output is as crucial as the output itself.

Our experiments will compare selected models on chart-based questions, focusing on whether their explanations are supported by the visual evidence. We will measure how often models provide a correct answer with an incorrect or unsupported explanation.

2 State of the Art

This section reviews 15 studies, beginning with the three closest to our project and then grouping the remaining works into three themes: chart evaluation, verifiable computation, and explanation reliability.

Table 1 summarizes the quantitative performance metrics reported in the recent literature.

2.1 Three closest studies

ChartQA-X, ChartHal, and FaithChart are the closest to our project because they address chart explanations, unsupported content, and visual grounding, respectively. Together, they provide useful datasets and evaluation methods, while leaving room for a more detailed analysis of individual explanation claims.

Chart explanation generation. ChartQA-X (Hegde et al., 2026) contains 30,799 samples pairing charts, questions, answers, and natural-language explanations across four chart types. The authors generate candidate explanations, select them using four criteria, and fine-tune models to produce answers with explanations. Overall, model explanations surpassed human-written ones in terms of these metrics. The paper also reports considerable improvements in question-answering accuracy and explanation-quality measure after fine-tuning. For our project, ChartQA-X provides explanation examples and a generation baseline.

Hallucination in chart understanding. ChartHal (Wang et al., 2025b) examines hallucination through 1,062 human-validated chart question-answering cases. It includes ordinary questions alongside prompts that refer to irrelevant, nonexistent, or contradictory chart information. On the full benchmark, the authors

Model / System	Dataset	Split / Subset	Metric	Score
GPT-5 (Wang et al., 2025b)	ChartHal	Full benchmark	Accuracy	34.46%
o4-mini (Wang et al., 2025b)	ChartHal	Full benchmark	Accuracy	22.79%
Claude 3.5 Sonnet (Masry et al., 2025)	ChartQA	Test set	Relaxed accuracy	90.50%
Claude 3.5 Sonnet (CoT) (Masry et al., 2025)	ChartQAPro	Full benchmark	Overall score	55.81%
GPT-4o (Wang et al., 2024)	CharXiv	Validation / reasoning	Accuracy	47.10%
Human baseline (Wang et al., 2024)	CharXiv	Validation / reasoning	Accuracy	80.50%
DePlot ensemble (1-shot) (Liu et al., 2023)	ChartQA	Test / human-written	Accuracy gain	+29.40 pp

Table 1: Published results on chart-understanding benchmarks. Results across different datasets and evaluation protocols are not directly comparable.

report accuracy of 34.46% for GPT-5 and 22.79% for o4-mini. Their analysis identifies absent and contradictory information as particularly challenging. These cases provide useful stress tests and error categories for our annotation guide.

Visual grounding of explanations. FaithChart (Mai et al., 2026), available as an SSRN preprint, directly studies whether chart explanations are grounded in visual evidence. It introduces sufficiency and comprehensiveness tests that modify cited image regions and measure how model predictions change. Its FaithChart-1500 benchmark contains 1,498 human-verified chart-explanation samples with reasoning traces. Authors suggest that a correct answer alone is insufficient evidence of a well-grounded explanation and report that their structured-evidence model reduces the gap between the two grounding metrics by 88%, with a 5.5-percentage-point reduction in accuracy.

2.2 Chart benchmarks and factual evaluation

Chart question-answering benchmarks provide useful evaluation data, but generally assess final answers rather than explanation quality. **ChartQA (Masry et al., 2022)** introduced roughly 32,700 questions and models using both chart images and underlying tables. Its released tables are particularly useful for checking values and arithmetic. **ChartQAPro (Masry et al., 2025)** extends evaluation to 1,948 questions about 1,341 charts from 99 sources, including dashboards, infographics, and unanswerable questions. **CharXiv (Wang et al., 2024)** adds 2,323 scientific charts with descriptive and reasoning questions, reporting accuracy of 47.1% for GPT-4o and 80.5% for humans.

Chart summarization research addresses factual support more directly. **ChartInsighter (Wang et al., 2025a)** uses 75 time-series charts with sentence-level error annotations and combines data-analysis modules with consistency checks. The authors report the lowest summary hallucination rate among their evaluated systems. **Obaid ul Islam et al. (Obaid Ul Islam et al., 2023)** show that training summaries can contain information absent from their chart inputs. Filtering these examples through natural language inference reduces hallucinations in human evaluation, with further gains from better chart-context encoding.

2.3 Separating chart reading from computation

Explicit intermediate representations can make chart reasoning easier to inspect. **DePlot (Liu et al., 2023)** converts chart images into tables before a language model answers the question, reporting a 29.4% improvement over its fine-tuned comparison model on ChartQA. **TinyChart (Zhang et al., 2024)** uses a 3-billion-parameter model to generate executable Python programs for numerical reasoning. Evaluated across five chart benchmarks, it reports stronger ChartQA performance than several larger chart models.

Faithful Chain-of-Thought Reasoning (Lyu et al., 2023) follows a related approach by translating questions into symbolic steps and using a deterministic solver to produce answers. It outperforms standard chain-of-thought on nine of ten benchmarks spanning mathematics, planning, question answering, and relational inference.

2.4 Reasoning explanations and verification

Prompting methods can improve answers without necessarily making explanations more reliable. **Chain-of-thought prompting (Wei et al., 2022)** uses worked examples with intermediate steps and reports substantial gains on arithmetic, commonsense, and symbolic tasks. **Self-consistency (Wang et al., 2023)** samples several reasoning paths and selects the most common an-

swer, improving accuracy over a single chain-of-thought sample.

Turpin et al. (Turpin et al., 2023) further show that plausible explanations may omit factors influencing model decisions. Across 13 BIG-Bench Hard tasks and a social-bias task, models often rationalize biased answers without acknowledging the biasing prompt features. **Chain-of-Verification (Dhuliawala et al., 2024)** offers a complementary method: draft an answer, generate factual checks, answer them independently, and revise the response. It reduces hallucinations on factual question-answering and long-form generation tasks.

3 Datasets

We've decided to focus on four particular datasets in our project. Let's evaluate these one by one.

3.1 ChartQA-X

This is a large-scale dataset built on ChartQA. It's been specifically designed to train multimodal transparency and reasoning quality.

- Status: `READY`;
- Size: 30k samples;
- Label Definitions: ground-truth answers paired with explanations;
- Class Balance: no distinct classes;
- License: CC-BY 4.0;
- Obtained: directly from Hugging Face `shamanthakhegde/ChartQA-X`;

3.2 ChartNet

This is a comprehensive dataset featuring over 1.5M chart samples, as well as other sub-datasets containing human-verified samples or question-answer pairs.

- Status: `READY`;
- Size: 1.5M+ samples;
- Label Definitions: 24 unique chart types;
- Class Balance: an uneven split of chart types ranging from 9% to less than 2%;
- License: Community Data License Agreement – Permissive, Version 2.0.;
- Obtained: directly from Hugging Face `ibm-granite/ChartNet`;

3.3 ChartHal

A dataset for comprehensive benchmarking of visual reasoning on charts.

- Status: `READY`;
- Size: 1000 samples;
- Label Definitions: four distinct question types, such as irrelevant or normal;
- Class Balance: an uneven four-way split;
- License: unknown, built upon CharXiv with Apache License 2.0;
- Obtained: cloned from project's huggingface site `hfl/ChartHal`;

3.4 ChartInsighter

A dataset specifically curated to evaluate hallucinations in time-series charts.

- Status: `READY`;
- Size: 2,000+ samples;
- Label Definitions: sentence-level hallucination labels divided into 10 error categories like range errors, omissions, numerical errors;
- License: CC-BY-4.0 license;
- Obtained: cloned directly from GitHub `wangfen01/ChartInsighter`;

4 Solution concept and experimental plan

For our experiments, we selected smaller, open-weight models that can be run locally. We hypothesize that their limited capacity may lead to more explanation errors, providing useful cases for analysis. These models also allow us to collect their generated reasoning traces.

In comparison, proprietary API models from providers such as OpenAI, Google and Anthropic expose reasoning summaries rather than full reasoning traces, which limits the material available for detailed analysis. Moreover, running open-weight models locally gives us greater control over model versions and settings, supporting reproducibility. If local computational resources prove insufficient, the same experiments can be hosted in the cloud.

We will compare two local multimodal models on chart images to measure both answer accuracy and the reliability of their explanations. The study will combine a manually annotated sample with a larger, automatically evaluated dataset and compare prompting strategies for reducing or detecting explanation errors.

4.1 Data gathering

We will construct two evaluation sets from the datasets described above: a small set for manual annotation, and a larger one for automatic annotation. An automated pipeline will run both selected

models and store their final answers, available generated reasoning outputs, and explanations. The models integrated into this pipeline will be:

- **Qwen3.5-9B** (Qwen Team, 2026),
- **Google/gemma-4-E4B-it** (Gemma Team, 2026).

Human-Annotated Dataset (200 Samples) We will select 25 chart-question pairs and evaluate two models using four strategies, producing 200 responses. All four team members will annotate the outputs for incorrect readings, invented information, unsupported claims, and calculation errors.

Large-Scale Auto-Annotated Dataset (1,000–10,000 Samples) To scale our investigation, we will select a larger sample of chart-question pairs from the datasets described above. After our two target models process these inputs, a stronger multimodal model, separate from those evaluated in the experiments, will serve as an *LLM-as-a-judge*. It will automatically annotate the generated explanations for hallucinations.

4.2 Prompting strategies

We will compare at least four of the following strategies, selected after assessing their suitability for chart-image inputs and the chosen models:

- **Direct prompting:** Ask each model for an answer and a brief explanation, without specifying intermediate steps.
- **Structured prompting:** Ask each model to list the relevant chart values before answering.
- **Chain-of-Thought:** Ask the model to describe the steps connecting the chart evidence to its answer.
- **Program-of-Thought:** Ask the model to generate executable code for numerical calculations and run it to obtain the result.
- **Self-consistency:** Generate multiple responses and select the most frequent normalized answer.
- **Verification pipeline:** Ask the model to check its answer and explanation against the chart, then revise them if needed.

4.3 Experimental plan

We will compare answer accuracy, explanation error rates, and the proportion of correct answers accompanied by flawed explanations. We will assess the automatic judge against human annotations using precision, recall, and F1, and report runtime, token usage, and API costs. The experimental pipeline is illustrated in Figure 1.

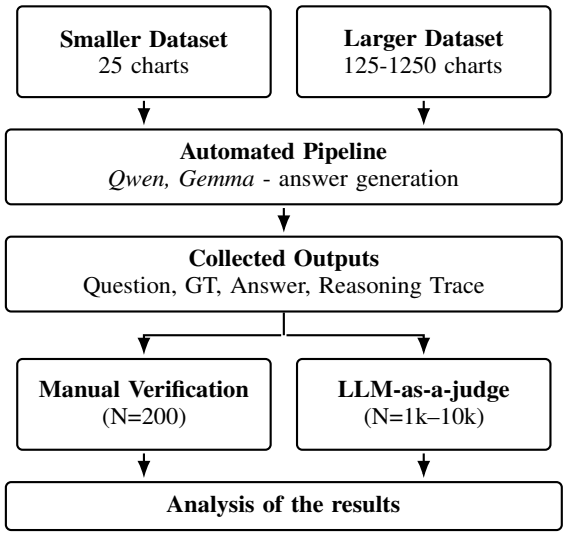


Figure 1: Methodology pipeline for detecting reasoning hallucinations.

4.4 Success deliverables

Depending on project progress and available resources, we aim to produce:

1. **Annotated explanations:** A manually annotated sample and a larger set annotated by a stronger LLM, both based on existing chart datasets and including correct answers with flawed explanations.
2. **Reproducible pipeline:** Scripts for local model inference, output collection, and automatic evaluation, together with prompting templates and model configurations.
3. **Comparative report:** An analysis of Qwen and Gemma across prompting strategies, covering answer accuracy, explanation errors, judge reliability, computational costs, and common failure patterns.

5 Disclosures

5.1 Data statement

Datasets listed in Section 3 are available on the Hugging Face platform. Licences for ChartQA - X, ChartNet and ChartInsighter allow for free use and modification. ChartHal does not have an explicitly stated license, but is built upon CharXiv, which uses an open-source Apache License 2.0. None of the datasets contain personal data.

Timeframe	Task / Milestone	Assignee(s)
Phase 1: Proposal (Deadline: Nov 4) – *Completed*		
Nov 4 – Nov 11	**PoC Setup:** Select 25 baseline charts; develop the automated Python pipeline for data collection.	Krzysztof, Tomasz, Piotr
Nov 11 – Nov 18	**PoC Execution:** Run baseline samples; manually annotate 200 reasoning traces; record mid-term video.	All (Annotation), Jan (Video prep)
Mid-term: Video & Presentation (Deadline: Nov 18)		
Nov 18 – Nov 25	**Scaling Up:** Run "LLM-as-a-judge" auto-annotation on 1k-10k samples.	Krzysztof, Piotr
Nov 25 – Dec 2	**Analysis:** Analyze common explanation errors; compare prompting strategies; prepare source code for submission.	Jan, Tomasz (Analysis), Krzysztof (Code)
Phase 2: Report, Source Code & Rebuttals (Deadline: Dec 2)		
Dec 2 – Dec 16	Address Phase 2 rebuttals	All
Dec 16 – Jan 13	Draft Phase 3 final report; record final video; prepare final presentation slides.	All
Phase 3: Final Presentation, Video & Report (Deadline: Jan 13, 2027)		

Table 2: Project timeline mapped to the official NLP course deadlines.

5.2 Contribution table

Task	P.J.	K.K.	J.W.	T.Ż.
Introduction			1h	
References	2h			
SOTA analysis	3h	0.5h	1h	
Datasets				4h
Experimental plan	2h	4h	1h	1h
Disclosures		0.5h	1h	

5.3 Project Timeline & Milestones

Table 2 outlines the task distribution and timeline for the entire project completion cycle.

5.4 LLM use

In this part of the project, we used GenAI tools to assist with literature searches, model selection, experimental planning, language correction, and LaTeX formatting.

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